CS5228-KDDM, 2025/26-2, Coursework 2

Introduction

- This coursework comprises two parts. Part 1 involves Python programming for regression and classification, and Part 2 contains four MCQs.
- Total CA marks of this coursework is 10. Details of marks/parts are below.
- A Canvas quiz will be open for your coursework submission.
- For Python programming parts, I urge you to complete a Jupyter notebook and submit it. Cw2-template.ipynb is the template for your answer. You have to run your codes and get sure that answers are available in the notebook before submission.
- Regarding MCQs, there is one and only one correct answer for each question. So select the best option. There is no penalty for wrong answers.
- The deadline for this coursework is **22/3/2026**. Please be aware that no delayed submission is possible.
- Good luck, my friends.

CW2, Part 1: Regression and Classification using Python (3+3= 6 marks)

CW2-1: Regression (3 marks)

Dataset: cs❶88❹_Housing¡csv

Goal: Estimation of house price

Algorithm: Linear Regression

Steps

1- The Dataset comprises information about different houses in America
2- We assume that the dataset is clean
3- The goal is to develop a linear regression model to estimate 'price' as a function of independent input variables { area, bedrooms, bathrooms, stories, mainroad, guestroom, basement, hot water heating, air conditioning, parking, prefarea, furnishing status}
4- Ignore "**class_label**" variable. This is the class label for the next question.
5- If you need to normalize your variables, apply that only to the input/independent variables.

6- If you need to convert your categorical data to numerical, either go for one-hot or unique integer coding. In the latter case, avoid coding from zero. Having many zeros will harm the linear regression model. Usually, however, it's not the case if you use one-hot.
7- You may apply a correlation analysis to see if any input variable is discardable.
8- Go for 90% training and 10% testing splitting.
9- Check if linear regression has got any hyperparameter that should be optimized or not.
10- Metrics are the mean square error (MSE) and mean absolute error (MAE) both reportable on both training and testing subsets.
11- Keep the splitting replicable. Use random_state=110 .
12- You cant change the model. However, using preprocessing, variable removal, and hyperparameters optimization, you may try to achieve the lowest possible testing MSE/MAE.
13- Using any typical Python library is allowed.

Deliverable

1- Use cw8_template¡ipynb as your template for both cw2-1 and cw2-2 questions.
2- Report if you have discarded any input variable and the justification for that, e.g., based on correlation analysis.
3- Print a few rows of your dataset along with the description of the dataset.
4- Print the size of training and testing subsets.
5- Print the MSE and MAE of the training phase
6- Print the MSE and MAE of the testing phase
7- If you applied any normalization, report the MSE and MAE with and without normalization, both training and testing.
8- Only the last/best MAE/MSE should be printed after doing whatever you did to obtain better results.

CW2-2: Classification using Decision Trees and Random Forest (3 marks)

Dataset:.cs❶88❹_Housing¡csv

Goal: Classification of houses into 4 price classes

Algorithm: Decision Tree and Random Forrest

Steps

1- The Dataset comprises information of different houses in America
2- We assume that the dataset is clean
3- The goal is to develop a classification model to classify the houses into 4 different price classes {cheap, medium, expensive, very expensive} based on features/input variables { area, bedrooms, bathrooms, stories, mainroad, guestroom, basement, hot water heating, air conditioning, parking, prefarea, furnishing status}
4- Ignore the "**price**" variable. This is the dependent variable employed for the regression part.
5- If you need to normalize your variables, do that.

6- If you need to convert your categorical data to numerical, either go for one-hot or unique integer coding.
7- You may apply a correlation analysis to see if any feature is discardable.
8- Go for 90% training and 10% testing splitting.
9- You will apply (train and test) decision tree and random forest models respectively. In both cases, hyperparameter optimization might be necessary.
10- Metrics are the Precision, Recall, and Accuracy (Pr/Rc/Acc), all reportable on both training and testing subsets.
11- Keep the splitting replicable. Use random_state=110 .
12- You can't change the models. However, using preprocessing, variable removal, and hyperparameters optimization, you may try to achieve the highest possible testing Pr, Rc, and Acc.
13- Using any typical Python library is allowed.

Deliverable

14- Use cw8_template¡ipynb as your template for both cw2-1 and cw2-2 questions.
15- Report if you have discarded any input variable and the justification for that, e.g., based on correlation analysis.
16- Print a few rows of your dataset along with the description of the dataset.
17- Print the size of training and testing subsets.
18- Print Pr/Rc/Acc of the training phase
19- Print Pr/Rc/Acc of the testing phase
20- If you applied any normalization, report the Pr/Rc/Acc with and without normalization, both training and testing.
21- Only the last/best Pr/Rc/Acc should be printed after doing whatever you did to obtain better results.
22- Print the confusion matrix, too.

CW2, Part 2: MCQs (4x1= 4 marks)

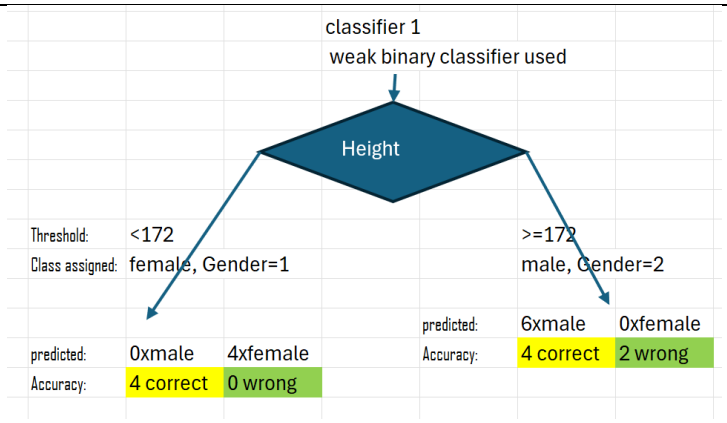
CW2-3- We are going to classify the labeled dataset below employing an Ada-Boost algorithm. We have 2 classes, Gender= Female, or 1, and Gender=Male or 2. Two features are extracted: Height and Weight. At the beginning of Ada-Boost algorithm, we have 2 weak binary classifiers, classifier 1 and 2. Their performance on the dataset and splitting threshold can be seen below, as well. Now please answer:

- What is the amount of say of each classifier?
- Rationally, which classifier will be used?
- What is the initial sample-weight array and the updated normalized sample-weight array after employing the classifier? w_i indicates the normalized sample-weight of i_{th} data sample.

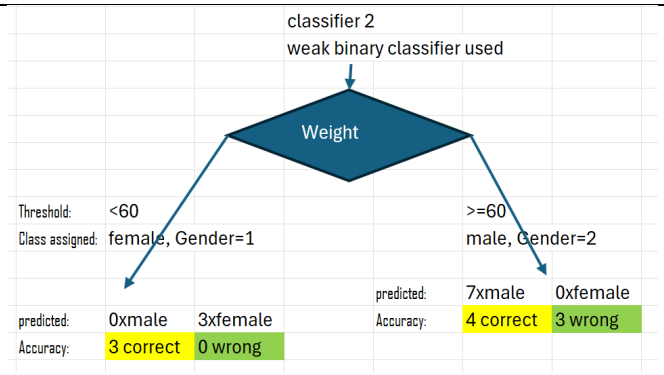
Celebrity	Height	Weight	Gender
Kylie Minogue	152	42	1
Jason Donovan	183	78	2
Elvis Presley	182	77	2
Kathy Parry	170	61	1
Sean Connery	188	75	2
Daniel Craig	178	78	2
Catherine Z-Jones	170	52	1
Sophia Loren	172.7	63.5	1
Ingrid Bergman	175.3	61.2	1
Madonna	161	45	1

1=female, 2=male

Dataset



Classifier 1



Classifier 2

a- Amount of say: classifier 1=0.5 , classifier 2=0.75, classifier 2 is selected. Normalized Sample-weights:

w_i	w_1	W_2	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}
Initial sample-weight	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11
Updated sample-weight	0.11	0.11	0.25	0.5	0.11	0.11	0.11	0.25	0.11	0.11

b- Amount of say: classifier 1=0.69 , classifier 2=0.42, classifier 1 is selected. Normalized Sample-weights:

w_i	w_1	W_2	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}
Initial sample-weight	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Updated sample-weight	0.063	0.063	0.063	0.063	0.063	0.063	0.063	0.25	0.25	0.063

c- Amount of say: classifier 1=0.69 , classifier 2=0.42, classifier 1 is selected. Normalized Sample-weights:

w_i	w_1	W_2	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}
Initial sample-weight	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1

Updated sample-weight	0.1	0.1	0.1	0.1	0.1	0.05	0.05	0.1	0.1	0.2

d- Amount of say: classifier 1=0.5 , classifier 2=0.75, classifier 2 is selected. Normalized Sample-weights:

w_i	w_1	W_2	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}
Initial sample-weight	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Updated sample-weight	0.25	0.063	0.063	0.063	0.063	0.063	0.063	0.063	0.063	0.25

CW2-4- In a classification task, the test dataset contains 50 data samples. There are 5 features, F1 to F5, and 4 classes, 1 to 4, in that test. The dataset is cw8_q0¡csv. We employed a decision tree classifier and came across the confusion matrix below. What are the macro precision (MAPr), and micro precision (MIPr) of that classifier?

		Actual			
Predicted	**Class**	**1**	**2**	**3**	**4**
	1	12	1	0	0
	2	1	11	2	0
	3	0	2	11	1
	4	0	0	2	7

a- MAPr= 0.82 , MIPr= 0.82
b- MAPr= 0.82 , MIPr= 0.78
c- MAPr= 0.78 , MIPr= 0.81
d- MAPr= 0.85 , MIPr= 0.85

CW2-5- Consider the **transaction database of 40 customers** in file cars9¡csv, where each transaction lists the car brands purchased by a customer over time. Assume we are applying the **Apriori algorithm** with:

- Minimum Support = 4 transactions
- Only pair itemsets (2-itemsets) are considered.

Which of the following **association rules has the highest confidence** among the candidates below?

a- Toyota → Nissan

b- Ford → Chevrolet

c- Hyundai → Kia

d- BMW → Audi

CW2-6- In **Gradient Boosting for regression**, assume we are minimizing the squared error loss, L:

$$L(y, F(x)) = (y - F(x))^2$$

At iteration m, the current model prediction is $F_{m-1}(x)$. The algorithm computes pseudo-residuals $r_i^{(m)}$for each training sample. Which of the following correctly describes these residuals? (6-1) Also, consider a Gradient Boosting model that sequentially adds weak learners (e.g., decision stumps) to minimize training error. Which of the following statements best explains why Gradient Boosting tends to reduce bias but may increase variance as the number of boosting iterations grows? (6-2)

a- 6-1: $r_i^{(m)} = y_i - F_{m-1}(x_i)$, 6-2: Because each new learner fits the residual errors of the current ensemble, the model becomes increasingly flexible and can capture complex patterns, which reduces bias but may cause sensitivity to training data fluctuations.

b- 6-1: $r_i^{(m)} = F_{m-1}(x_i) - y_i$, 6-2: Because each weak learner is trained independently on random subsets of data, the ensemble becomes more stable and therefore variance increases.

c- 6-1: $r_i^{(m)} = \frac{\partial L(y_i,\ F(x_i))}{\partial y_i}$, 6-2: Because Gradient Boosting forces all weak learners to share the same parameters, the model complexity increases and bias decreases.

d- 6-1: $r_i^{(m)} = \frac{\partial L(y_i,\ F(x_i))}{\partial F_m(x_i)}$, 6-2: Because boosting replaces the original training labels with predicted labels, the ensemble eventually memorizes the data.

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